Besides the water-clock, Isaac made a sun-dial. Thus his grandmother was never at a loss to know the hour; for the water-clock would tell it in the shade, and the dial in the sunshine. The sun-dial is said to be still in existence at Woolsthorpe, on the corner of the house where Isaac dwelt.

Isaac possessed a wonderful faculty of **acquiring** knowledge by the simplest means. For instance, what method do you suppose he took, to find out the strength of the wind?. He jumped against the wind; and by the length of his jump, he could calculate the force of a gentle breeze, a brisk gale, or a tempest. Thus, even in his boyish sports, he was continually searching out the secrets of philosophy.

Not far from his grandmother's residence there was a windmill, which operated on a new plan. Isaac was in the habit of going thither frequently, and would spend whole hours in examining its various parts. While the mill was at rest, he **pried** into its internal machinery. When its broad sails were set in motion by the wind, he watched the process by which the mill-stones were made to revolve, and crush the grain that was put into the **hopper**. After gaining a thorough knowledge of its construction, he was observed to be unusually busy with his tools.

It was not long before his grandmother, and all the neighborhood, knew what Isaac had been about. He had constructed a model of the windmill. Though not so large, I suppose as one of the box-traps which boys set to catch squirrels, yet every part of the mill and its machinery was complete. Its little sails were neatly made of linen, and whirled round very swiftly when the mill was placed in a draught of air. Even a puff of wind from Isaac's mouth, or from a pair of bellows, was sufficient to set the sails in motion. And what was most curious if a handful of grains of wheat were put into the little hopper, they would soon be converted into snow-white flour.



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